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Comprehensive utilization of lignocellulosic biomass for the electrode and electrolyte in zinc-ion hybrid supercapacitors†

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Zinc-ion hybrid supercapacitors (ZHSs) with carbon cathodes and aqueous electrolytes are promising as they easily combine advantages such as high energy density, high power density, long cycle life and good safety. Herein, a novel strategy can simultaneously fabricate porous carbon as the cathode and a cellulose-based hydrogel as the electrolyte from biomass. Lignocellulosic biomass is hydrothermally treated with a concentrated $ZnCl_2$ solution to obtain a solid residue and a filtered liquid. From the solid residue, porous carbon can be directly prepared by carbonization and it shows a high specific capacitance of 263 F g^{-1} for the storage of zinc salt ions. From the filtered liquid, the cellulose-based hydrogel is synthesized by the addition of an ionic crosslinking agent and it has a wide voltage window of 2.0 V as the electrolyte for ZHSs. The ZHSs assembled from the sustainable carbon cathode and hydrogel electrolyte show a high capacity of 229 mA h g^{-1} at 0.5 A g^{-1} and retain a capacity of 86 mA h g^{-1} at 10 A g^{-1}, and the energy density reaches 226 W h kg^{-1} with a power density of 492 W kg^{-1}. The ZHSs also have an excellent capacity retention of almost 100% after 30 000 cycles.

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Introduction

Inefficient use of natural resources has brought about resource shortages and environmental degradation.[1] Traditional energy sources are no longer able to meet the needs of human production and life in the developed economic society. Developing a green, sustainable, efficient, and widely applicable energy storage technology/device is particularly urgent for energy utilization and development.[2] As a new electrochemical energy storage device, zinc-ion hybrid supercapacitors (ZHSs) have attracted great attention due to their satisfactory energy/power density, long cycle life, and good safety as well as low manufacture cost.[3,4] Generally, ZHSs can be assembled from a zinc ion-containing electrolyte, a capacitor-type cathode and a zinc metal anode.[5–7] The energy storage mechanism of ZHSs is based on the reversible deposition/stripping of zinc ions on the anode and the reversible adsorption/desorption of ions on the cathode.[8,9] Compared with zinc deposition/stripping, the ion adsorption/desorption is restricted due to the limited adsorption sites.[10,11] The cathode and electrolyte are the keys to the energy storage of ZHSs with a high capacity.[12,13]

For the cathode of ZHSs, carbon materials are ideal candidates due to their stable physicochemical properties, accessible high specific surface area, and good electronic conductivity as well as abundant sources.[14,15] In particular, porous carbon with abundant mesoporous and macroporous structures is highly desirable for the storage and transport of zinc salt ions.[16,17] The zinc salt ions have big ion diameters (*e.g.*, 0.86 nm for hydrated Zn^{2+}, 0.66 nm for hydrated Cl^- and 0.76 nm for hydrated SO_4^{2-}), and the cathode of ZHSs needs to have enough space to handle the exchange adsorption of anions and cations.[18] For the electrolyte, hydrogels with high ionic conductivity and excellent safety could be easily assembled into ZHSs and also avoid liquid leakage.[19] Aqueous hydrogel electrolytes are expected to play a vital role in wearable/flexible energy storage devices.[20,21] However, the hydrogel electrolytes for ZHSs are mostly fabricated from artificial polymers, and more consideration should be given to natural polymers to achieve renewable hydrogels.[21,22] Furthermore, ZHSs with aqueous electrolytes normally show a limited stable electrochemical window due to the voltage intolerant water.[23,24] In most of the previous work, the electrode and electrolyte materials were separately produced from different sources and methods.[20,25–29]

Lignocellulosic biomass, with the main components cellulose, lignin and hemicellulose, is one of the most abundant renewable resources in the world.[30–32] Lignocellulosic biomass is a good source for fabricating carbon materials with abundant oxygen functional groups that can promote the storage of zinc ions.[33,34] Additionally, the cellulose component in lignocellulosic biomass is a potential natural polymer for the synthesis of hydrogels.[35,36] Here, we introduced a new strategy to prepare porous carbon and cellulose-based hydrogels *via* the

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comprehensive utilization of lignocellulosic biomass. $ZnCl_2$ was used for the dissolution of biomass components, and also as a molten salt agent for the carbonization and activation of biomass sources.[37–40] When lignocellulosic biomass was pretreated with a $ZnCl_2$ aqueous solution, its components were separated into a solid residue and a dissolved substance. The solid residue was carbonized into porous carbon, and the filtered liquid was coagulated to form a cellulose-based hydrogel. This strategy can simultaneously fabricate the carbon cathode and hydrogel electrolyte with excellent properties for ZHSs. The assembled ZHSs show a wide voltage window and a long cycle life, and achieve a high capacity of 229 mA h g^{-1} and an energy density of 226 W h kg^{-1} at a power density of 429 W kg^{-1}.

Experimental section

Preparation of lignocellulosic biomass-derived porous carbon and hydrogels

30 g of $ZnCl_2$ was mixed with 10.8 g of H_2O to obtain a clear solution with a $ZnCl_2$ concentration of 73 wt%. 1.5 g of wood sawdust (pine, *Pinus* species) was added into the $ZnCl_2$ solution for hydrothermal treatment with stirring at 75 °C for 1, 3 and 6 h. Then, the resulting suspensions were separated into filtered liquids and solid residues. The separation was controlled by suction filtration with a 200 mesh filter cloth until there were no droplets. The unwashed solid residues were directly used for carbonization by heating to 600 °C for 2 h under a nitrogen atmosphere with a heating rate of 10 °C min^{-1}. After that, the carbonized products were washed with water and HCl (5 wt%) to obtain lignocellulosic biomass-derived carbon (LDCx, $x = 1$, 3 and 6, standing for the hydrothermal treatment times of 1, 3 and 6 h, respectively). The washed solid residues (WSRx) with different hydrothermal treatment times of 1, 3 and 6 h (WSR1, WSR3 and WSR6) were also collected. For the filtered liquids, a specific amount of $CaCl_2$ was added with a mass ratio of filtered liquid to $CaCl_2$ of 20 : 1 at 75 °C. After cooling the viscous liquids, lignocellulosic biomass-derived hydrogels (LDHx) with different hydrothermal treatment times of 1, 3 and 6 h (LDH1, LDH3 and LDH6) were obtained.

Assembly of zinc-ion hybrid supercapacitors

The prepared powdered carbon (LDCx) was mixed with Super P carbon black and polyvinylidene fluoride (PVDF) in *N*-methylpyrrolidone (NMP) to form a slurry. The mass ratio of LDCx, Super P carbon black and PVDF was 8 : 1 : 1, and the PVDF was pre-dispersed in NMP with a mass fraction of 5%. The slurry was evenly coated on a titanium mesh with a mass loading of around 8.0 mg cm^{-2}, and dried to obtain the LDCx cathode electrode. A nonwoven fabric soaked with aqueous $ZnCl_2$ solution (73 wt%) was used as the electrolyte and separator to assemble the ZHSs with the LDCx cathode and a zinc foil anode. The prepared hydrogel LDHx was used as a solid-state electrolyte to fabricate ZHSs with the LDCx cathode. For comparison, a nonwoven fabric soaked with an aqueous $ZnCl_2/CaCl_2$ mixture solution (69.5 wt% $ZnCl_2$, 4.8 wt% $CaCl_2$ and 25.7 wt% H_2O) was also used as the electrolyte for testing.

Characterization

Fourier transform infrared spectroscopy (FTIR, Thermo Electron Nicolet-360), X-ray diffraction (XRD, Rigaku MiniFlex II) and scanning electron microscopy (SEM, JSM-7600F, JEOL Ltd.) were used to analyze the solid samples. The carbon samples were further analyzed by X-ray photoelectron spectroscopy (XPS, Daojin AXIS UltraDLD) and N_2 adsorption–desorption isotherms (Micromeritics TriStar II 3020). The chemical component analysis was conducted according to the National Renewable Energy Laboratory procedures.[41] Briefly, the cellulose and hemicellulose contents of the raw material were calculated from the monomeric sugar concentration after sulfuric acid hydrolysis, and then the lignin content was measured by gravimetric analysis. The tensile tests of hydrogel samples were conducted using a universal testing machine (CMT850) at a stretching rate of 30 mm min^{-1}.

Electrochemical measurements

The ionic conductivity (σ, mS cm^{-1}) of the hydrogels was tested by electrochemical impedance spectroscopy (EIS). Two stainless steel sheets were used as the electrodes to clamp the hydrogels for the test. The ionic conductivity was calculated using the equation $\sigma = 1000L/(AR)$, where L (cm) is the thickness of the hydrogel, A (cm^2) is the effective contact area of the hydrogel and electrode and R (Ω) is the intercept of the Nyquist plot at the real axis. For zinc-ion hybrid supercapacitors, the cyclic voltammetry (CV) and galvanostatic charge–discharge (GCD) were tested. The specific capacitance (C_f, F g^{-1}) was calculated using the equation $C_f = (I/m) \times \Delta t/\Delta V$, where I/m is the current density (A g^{-1}) according to the mass of carbon material on the cathode, ΔV is the discharge voltage window (V) and Δt is the discharge time (s). The energy density (E, W h kg^{-1}) was computed by $E = C_m \times \Delta V/2$, and C_m is the specific capacity (mA h g^{-1}) which can be calculated by $(I/m) \times \Delta t/3.6$. The power density ($P$, W kg^{-1}) was calculated by $P = E/\Delta t \times 3600$. An electrochemical work station (CHI 760e, Chenhua, Shanghai) was used to conduct the electrochemical measurements except for long-term GCD which was conducted using a Land test system (CT2001A, Wuhan China).

Results and discussion

Synthesis route of lignocellulosic biomass-derived carbon and hydrogels

As shown in Fig. 1, wood sawdust with the components cellulose, hemicellulose and lignin is hydrothermally treated with a concentrated $ZnCl_2$ aqueous solution. The highly concentrated $ZnCl_2$ with a specific hydrate composition of $[Zn(OH_2)_6][ZnCl_4]$ can donate hydrogen bonds to trap the hydroxyl groups, and thus make a "soluble" cellulose chain.[39] After separation by filtration, a solid residue and a cellulose-containing $ZnCl_2$ solution (filtered liquid) can be obtained. The unwashed solid residue is good for the synthesis of porous carbon. The filtered liquid with dissolved cellulose and abundant zinc salt ions can be coagulated into cellulose-based hydrogels using a Ca^{2+} ionic crosslinking agent.[36] The addition of Ca^{2+} helped form the homogeneous gelling solution.[40]

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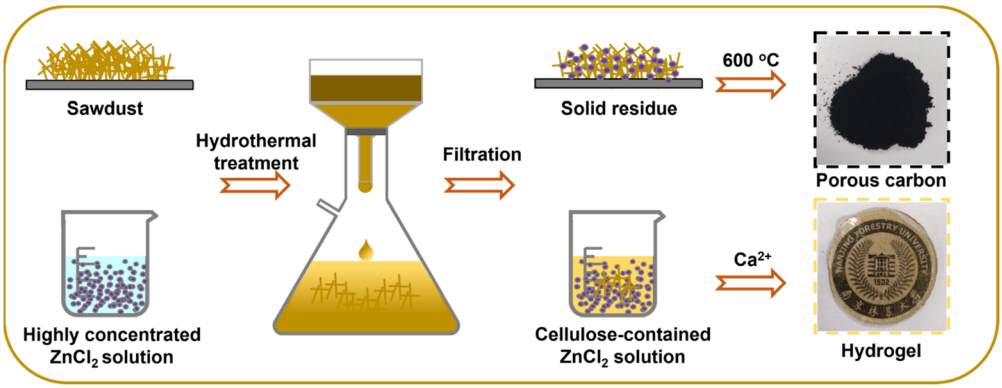

Fig. 1 Synthesis route of porous carbon and hydrogels from lignocellulosic biomass.

Hydrothermal treatment of lignocellulosic biomass

The chemical components of raw sawdust and solid residues (WSR*x*) with different hydrothermal treatment times were analyzed (Fig. 2a). The solid yields of WSR*x* decrease with the increase in hydrothermal treatment time (75% for WSR1, 60% for WSR3 and 29% for WSR6). Compared with the raw sawdust (41.3% cellulose, 17.2% hemicellulose and 34.6% lignin), WSR1 presents decreased cellulose (30.5%) and hemicellulose content (8.5%), whereas the lignin content almost does not change (34.0%). When the hydrothermal treatment time increased to 6 h, the cellulose, hemicellulose and lignin content decreased to 13.2%, 1.6% and 13.7% respectively (WSR6). The detailed components of WSR*x* are shown in Table S1.† The cellulose and hemicellulose components dissolved faster than the lignin component, thus resulting in the increased content of lignin in solid residues. Fig. 2b shows the FTIR spectra of raw sawdust and WSR*x*. Specifically, the peak at 1730 cm^{-1} is mainly attributed to the vibration of C=O of carboxylic acids, and the peak at 1590 cm^{-1} represents the vibration of the aromatic skeleton.[42] The WSR*x* show enhanced vibration peaks at 1590 cm^{-1} relative to 1730 cm^{-1} compared with the raw sawdust, corresponding to the increased content of lignin. The WSR*x* retain higher lignin content than cellulose content, indicating the selective ability of $ZnCl_2$ in the dissolution of cellulose from the raw sawdust, especially at short hydrothermal treatment time.

Fig. 2c shows the XRD patterns of sawdust and WSR*x*. For the raw sawdust, two diffraction peaks at 16.4° and 22.5° can be observed, which correspond to the (101) and (002) planes of I-type cellulose. After hydrothermal treatment, WSR*x* present weakened diffraction peaks shifting towards lower angle,

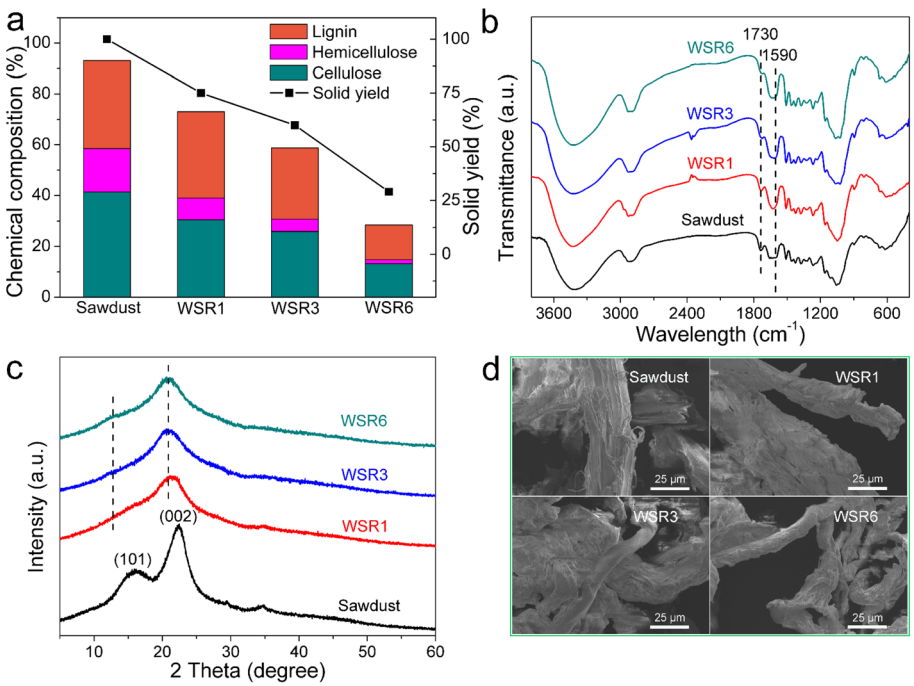

Fig. 2 Solid yields and chemical components (a), FTIR spectra (b), XRD patterns (c) and SEM images (d) of sawdust and WSR*x*.

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indicating the formation of II-type cellulose.[43] According to the SEM images, the cellulose fiber bundle in the raw sawdust was changed into individual swollen fibers that can be well identified in WSR*x* (Fig. 2d). During the treatment with $ZnCl_2$ solution, the cellulose crystallinity and lignin structure were gradually broken.[32]

Porous carbon derived from solid residues

The solid residues were directly calcined without removing $ZnCl_2$ to produce porous carbon LDC*x*. The XRD patterns of LDC*x* show two broad diffraction peaks for the (002) and (100) planes at 23.9° and 42.8°, indicating limited graphitization at a low carbonization temperature (600 °C) (Fig. 3a).[34] The FTIR spectra of LDC*x* present the bands of oxygen-containing groups inherited from the raw sawdust, and the peaks at 3430, 1590 and 1160 cm^{-1} represent the vibrations of O–H, C=O and C–O, respectively (Fig. 3b).[42] The O heteroatoms in the porous carbon were further detected by XPS survey (Fig. S1†). The O 1s spectrum can be divided into three main peaks at around 531.4, 532.5 and 533.8 eV which correspond to the functional groups of –C=O, –C–OH and –COOH, respectively (Fig. 3c).[9] The –C–OH and –COOH on the porous carbon can participate in the storage of Zn^{2+} by the specific formation of –C–O–Zn and –COO–Zn, and the carbonyl –C=O can greatly enhance the wettability of the carbon cathode by the aqueous electrolyte.[11,26,34] The widespread O element in the porous carbon can also be observed using a combination of SEM and EDS mapping images (Fig. S2†).

The carbonization temperature of 600 °C is carefully chosen for the $ZnCl_2$ activation of solid residues.[38,44] During the carbonization, the $ZnCl_2$ inside the solid residues also functions as a molten salt template with fluidity for the formation of the porous structure. The LDC*x* are composed of aggregated nanoparticles, where the nanoparticles in LDC1 and LDC3 are tightly packed (Fig. 3d–f). The N_2 adsorption–desorption isotherms of LDC*x* were recorded, and all the samples present combined isotherms of II and IV types, suggesting that the structure has abundant mesopores (Fig. 3g).[45] Barrett–Joyner–Halenda analysis demonstrates a slight increase in pore size from LDC1 to LDC6 (Fig. S3†). The lignin in the raw material was gradually destroyed with increasing the hydrothermal treatment time from 1 to 6 h. The carbon precursors with fragmented lignin and weakly crystalline cellulose are easily decomposed and dispersed in the $ZnCl_2$ molten salt during carbonization, leading to the formation of porous carbon materials. Samples LDC1, LDC3 and LDC6 exhibit high Brunauer–Emmett–Teller (BET) surface areas of 1618, 1576 and 1511 $m^2\ g^{-1}$, and the slight reduction in BET surface area from LDC1 to LDC6 corresponds to the increase in pore size.

As shown in Fig. 3h, the capacitance performance of LDC*x* as the cathode for ZHSs was studied. At a low current density of 0.5 A g^{-1}, LDC1, LDC3 and LDC6 have the specific capacitances of 263, 246 and 238 F g^{-1}, respectively. As the current density increases to 10 A g^{-1}, LDC1, LDC3 and LDC6 show specific capacitances of 172, 170 and 139 F g^{-1}, respectively (the complete GCD curves at different current densities can be found

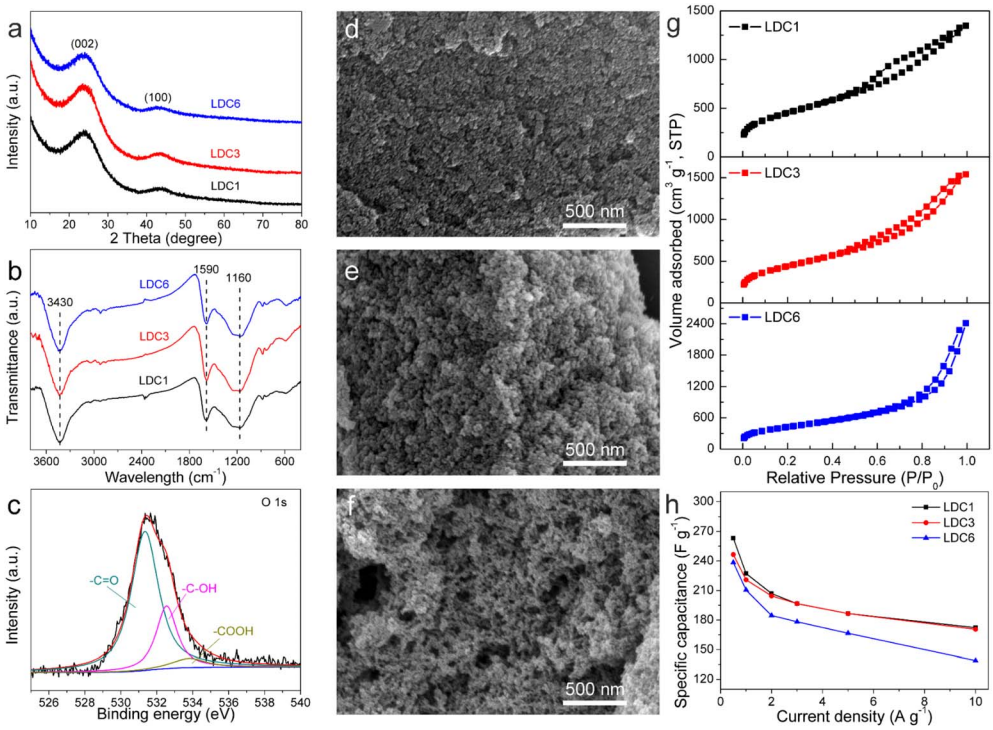

Fig. 3 XRD patterns (a) and FTIR spectra (b) of carbon LDC*x*. O 1s XPS spectrum (c) of LDC3. SEM images of LDC1 (d), LDC3 (e) and LDC6 (f). N_2 adsorption–desorption isotherms of LDC*x* (g). Rate capability of LDC*x* with a concentrated aqueous $ZnCl_2$ solution as the electrolyte in a zinc-ion hybrid supercapacitor (h).

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in Fig. S4†). As reported in our previous work, a micropore-dominated activated carbon with abundant oxygen functional groups and high BET surface area (1522 m^2 g^{-1}) only exhibits a specific capacitance of 93 F g^{-1} at 10 A g^{-1} with a $ZnCl_2$ electrolyte.[45] A carbon material with abundant mesopores is more suitable for storing zinc salt ions. LDC1 and LDC3 with tightly packed nanoparticles show better rate capability than LDC6. Compared with LDC1 and LDC3, LDC6 presents a similar CV curve at a low scan rate of 5 mV s^{-1}, whereas it exhibits an incongruent shape at a high scan rate of 200 mV s^{-1} (Fig. S5†). For the preparation of carbon materials from lignocellulosic biomass, it is desirable to preserve the lignin structure within the carbon precursor to obtain porous carbon with a suitable pore structure.

Hydrogels derived from filtered liquids

The filtered liquids with highly concentrated $ZnCl_2$ after the hydrothermal process were directly used to synthesize quasi-solid-state hydrogel electrolytes (LDH1, LDH3 and LDH6) by crosslinking the dissolved cellulose with a small amount of Ca^{2+}. According to tensile tests (Fig. 4a), the hydrogel exhibits increasing tensile stress from 21.1 kPa (LDH1) to 52.4 kPa (LDH6) with increasing hydrothermal treatment time, which was caused by the increased amount of dissolved cellulose. Among the hydrogels, LDH3 has the best stretchability of 46.5%. The inferior stretchability of LDH6 compared to LDH3 is attributed to excessive dissolved lignin. The ionic conductivity of LDH*x* was tested by EIS (Fig. 4b). LDH1, LDH3 and LDH6 have ionic conductivity of 22.25, 16.22 and 7.94 mS cm^{-1}, respectively. Lignin in the liquid phase results in poor ionic conductivity in the final hydrogel electrolyte. As shown in Fig. 4c, the hydrogel becomes darker in colour from LDH1 to LDH6, corresponding to the increased lignin content. Due to the high inorganic salt concentration, the hydrogel remains flexible at low temperature (Fig. 4d and e), and has ionic conductivities of 11.49 and 7.05 mS cm^{-1} at 0 and −20 °C (LDH3, Fig. S6†). Additionally, the large amount of $ZnCl_2$ with water retention ability also makes the hydrogel resistant to drought.

Considering the mechanical properties and ionic conductivity, LDH3 was selected to investigate the ion storage in hybrid supercapacitor. The corresponding LDC3 was used as the cathode to assemble ZHSs with LDH3. Controllable voltage windows are explored with the CV curves for different voltage ranges (Fig. 4f). The symmetrical quasi-rectangular shape of the CV curves can be observed even in a wide voltage range beyond 2.0 V. The galvanostatic charge and discharge of the ZHSs were also tested under different defined voltage windows (Fig. S7†). When the voltage exceeds 2.0 V by charging, a voltage plateau begins to appear. The voltage plateau above 2.0 V (*vs.* Zn/Zn^{2+}) is mainly attributed to the oxidation of chloride ions at the cathode, which has some reversibility during discharge.[10,46] Benefiting from the hydrogel electrolyte, the ZHSs have a wide voltage window. The hydrogel electrolyte was compared with liquid aqueous electrolytes (concentrated $ZnCl_2$ solution and $ZnCl_2/CaCl_2$ mixture solution) by CV tests (Fig. 4g). The $CaCl_2$ in the mixture solution has the same mass fraction as that in the hydrogel. LDH3 shows a CV shape similar to that of $ZnCl_2$ solution rather than $ZnCl_2/CaCl_2$ mixture solution. For the hydrogel electrolyte, the Ca^{2+} ions as crosslinkers are immobilized and only the ion species of $ZnCl_2$ participate in energy storage as charge carriers.[36] The mobilizable Ca^{2+} in the liquid $ZnCl_2/CaCl_2$ mixture solution may intervene in the diffusion of Zn^{2+}, resulting in the poor redox peaks in the CV curve.

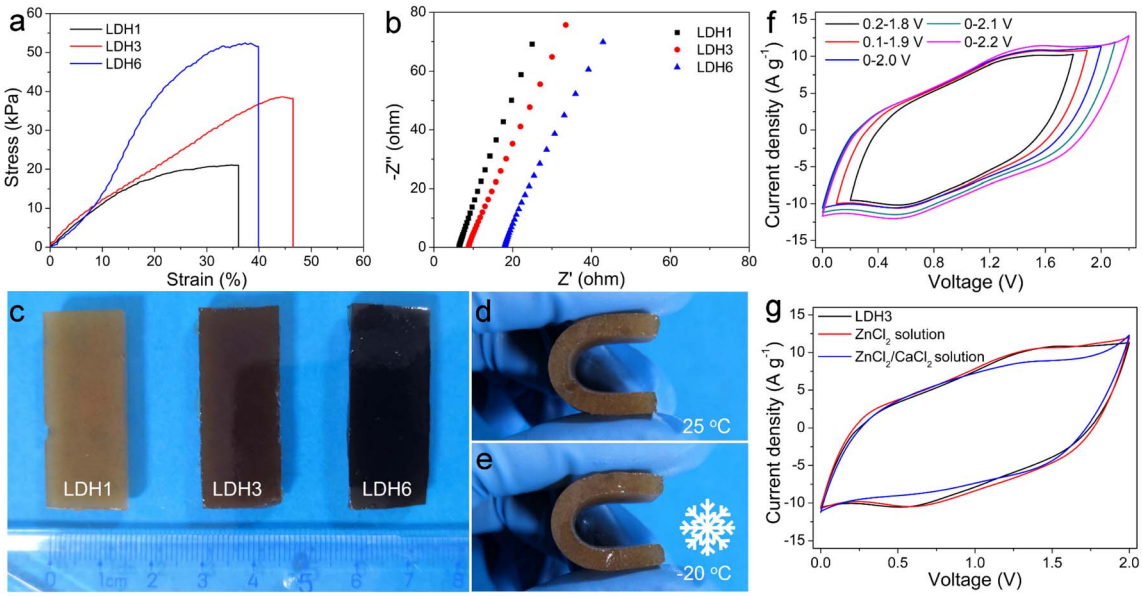

Fig. 4 Tensile stress–strain curves (a) and Nyquist plots (b) of LDH*x*. Digital images of hydrogel LDH*x* (c) and the demonstration of flexibility at 25 (d) and −20 °C (e). CV curves of ZHSs assembled from the LDH3 electrolyte and LDC3 carbon cathode at 50 mV s^{-1} with different voltage windows (f). CV curves at 50 mV s^{-1} for the ZHSs with different electrolytes (g).

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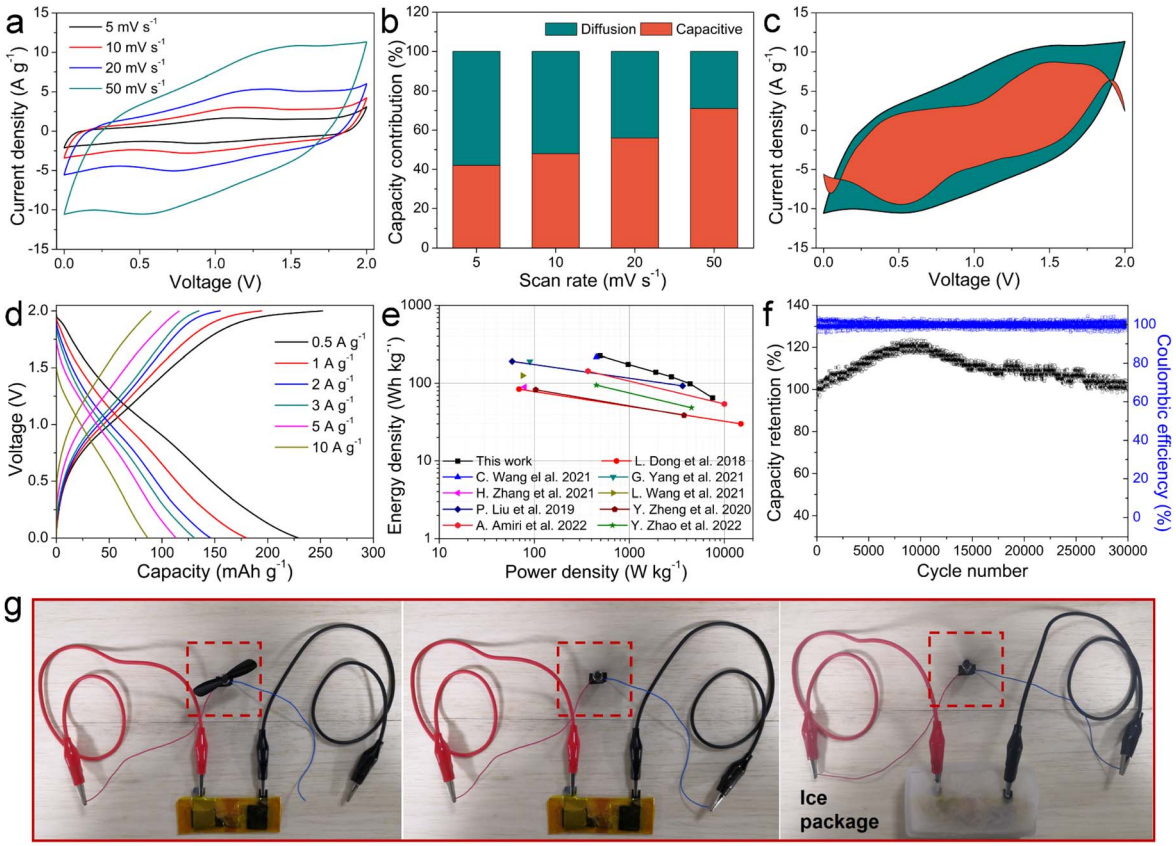

Fig. 5 ZHSs with lignocellulosic biomass-derived carbon cathode and hydrogel electrolyte: CV curves at different scan rates (a), capacitive contributions at 50 mV s^{-1} (b), normalized capacity contributions at different scan rates (c), GCD curves at different current densities (d), Ragone plots (e), GCD cycles at 5 A g^{-1} (f), and application demonstration (g).

Capacitive performance of ZHSs assembled from lignocellulosic biomass-derived carbon and hydrogel

Within the stable voltage window of 2.0 V, the electrochemical performance of ZHSs (Zn/LDH3/LDC3) was further studied. The energy storage of ZHSs is contributed by both capacitive and diffusion behaviors. The CV curves of the ZHSs present quasi-rectangular shapes with redox peaks at different scan rates (Fig. 5a). The response currents of CV can be quantified using the equation $i = i_{cap} + i_{diff} = k_1\nu + k_2\nu^{1/2}$, where ν is the scan rate, and $k_1\nu$ and $k_2\nu^{1/2}$ represent the capacitive current and diffusion current, respectively. According to the calculation, the contribution of the capacitive process to the capacity gradually increases from 42% at 5 mV s^{-1} to 71% at 50 mV s^{-1}, indicating relatively rapid electrochemical kinetics at high scan rates (Fig. 5b and c). GCD tests were conducted at different current densities (Fig. 5d). As a hybrid energy storage device, the ZHSs reach a "battery level capacity" of 229 mA h g^{-1} at 0.5 A g^{-1} and retain a capacity of 86 mA h g^{-1} at a high current density of 10 A g^{-1}. The carbon material with abundant mesopores provides a high capacity for the storage of zinc salt ions.[16] Furthermore, benefiting from the high concentration and easy desolvation process of $ZnCl_2$ species, the hydrogel electrolyte can balance the kinetics between Zn deposition/stripping at the anode and ion adsorption/desorption at the cathode, thus resulting in good rate capability.[5,47] Combining wide voltage window, high capacity and good rate capability, the ZHSs can achieve high energy density of 226 W h kg^{-1} at a power density of 492 W kg^{-1} and 65 W h kg^{-1} at 7515 W kg^{-1}, and outperform most previously reported zinc-ion hybrid supercapacitors (Fig. 5e).[3,5,7,12,25,26,28,34,48] The cycling stability was tested at a moderate current density of 5 A g^{-1} for charging and discharging (Fig. 5f). The coulombic efficiency remains at around 100% throughout all the cycles. After 30 000 cycles, the capacity remains almost 100% with respect to the original cycle. At a low current density of 0.5 A g^{-1}, the ZHS is also stable in long-term cycling (Fig. S8†). A comprehensive comparison shows that our ZHS is competitive with other ZHSs (Table S2†). Fig. 5g demonstrates the application of the ZHSs. A 3.7 V motor could be powerfully driven by two ZHSs in series, and the ZHSs still worked well with an ice package (Video S1†).

Conclusions

In summary, we have demonstrated a novel strategy to simultaneously fabricate porous carbon and hydrogels from lignocellulosic biomass using $ZnCl_2$. The solid residues after the hydrothermal process can be directly carbonized to prepare

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porous carbon with abundant mesopores and oxygen functional groups. The filtered liquid after the hydrothermal process can be coagulated into a zinc ion-containing hydrogel by adding an ionic crosslinking agent. The porous carbon affords a high capacitance of 263 F g^{-1} for the storage of zinc salt ions. The hydrogel provides the zinc-ion hybrid supercapacitors with a stable operating voltage of 2.0 V as well as a good mobility of zinc salt ions. The ZHSs assembled from the lignocellulosic biomass-derived carbon cathode and hydrogel electrolyte achieve a high energy density of 226 W h kg^{-1} and a high power density of 7515 W kg^{-1}. The ZHSs also have an excellent cycle life with 100% capacity retention after 30 000 cycles. We believe that this strategy to prepare sustainable porous carbon and hydrogels is applicable to all lignocellulosic biomass and will facilitate the practical applications of zinc-ion hybrid energy storage devices.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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